

Retail Demand for a Counter-Inflation Currency: A Simulation Under Explicit Assumptions

Category One Limited

Model version 3 • Draft

Abstract

This study simulates retail adoption of a monetary instrument offering protection against loss of purchasing power. It asks what share of household savings would move into such an instrument over ten years, given stated assumptions about how households learn of it, gain access to it, and come to trust it. The instrument’s protective property is taken as an input and is not tested. Seventeen behavioural parameters with no direct measurement were sampled simultaneously from continuous ranges across 8,000 draws, split between a developed-market and a high-inflation specification. Adoption occurred in 99.92 per cent of developed-market samples and in all high-inflation samples. Median ten-year outcomes were 13.2 and 12.5 per cent of household savings balances respectively, with fifth percentiles of 1.1 and 3.9 per cent. The binding constraint differs by market: trust in developed economies, awareness where inflation is high. No dataset exists on general-public demand for an instrument of this kind, and the study does not constitute one.

1 Scope

This is a simulation of retail adoption. It does not predict adoption; it shows what follows from its assumptions. It does not validate the instrument: full purchasing-power protection enters as a parameter $\pi = 1$, taken as given per the system’s specification, and every result is conditional on that holding. It does not measure demand; no dataset on general-public demand for an inflation-protected savings instrument exists, and every available source either samples existing cryptocurrency users or measures a different product. It does not model commercial or regulatory feasibility, with the exception noted in Section 7.

The central result is not a point estimate. It is that adoption occurs across essentially the entire sampled parameter space, including at its pessimistic extremes.

2 Model

An agent-based discrete-time model over monthly steps $t = 0, \dots, T$ with $T = 120$, run over populations of n synthetic households per market.

Each household i carries a savings balance b_i drawn lognormally, a current holding $h_i \in \{L, S, O\}$ denoting local currency, US-dollar stablecoin, or other store of value, an annual savings-drawdown fraction d_i , a trust threshold θ_i , and binary states for awareness and account access.

2.1 Marginal gain

The annual benefit from adoption depends on current holding, not on location:

$$g_i = \pi \cdot \begin{cases} \iota_L & h_i = L \\ \iota_U + \rho\sigma & h_i = S \\ \frac{1}{2}\iota_U + \rho\sigma & h_i = O \end{cases} \quad (1)$$

where ι_L is local inflation, ι_U US inflation, ρ the perceived annual probability of a dollar stress event and σ its severity.

This asymmetry is the model's central economic mechanic. A household holding Argentine pesos gains approximately 0.30; one already holding a dollar-denominated stablecoin gains approximately 0.03, identically in Buenos Aires and in London. Treating these as one population is the principal source of overstatement in conventional total-addressable-market estimates.

2.2 Access

Day-one access is restricted to households already holding a cryptocurrency exchange account. In-app threshold-signature wallets combined with decentralised-exchange liquidity render purchase a two-step operation for that group, requiring no listing decision by any venue. Households without an account must first open one and complete identity verification.

Account penetration grows toward a ceiling:

$$a(t) = \min\left(a_0(1 + \gamma)^{t/12}, \bar{a}\right) \quad (2)$$

with a_0 current penetration, γ annual growth and $\bar{a}$ the long-run ceiling. Household i has access at t if $u_i < a(t)$, where $u_i \sim U(0, 1)$ is drawn once.

2.3 Salience

Protection-motivated switching is rapid when a threat is vivid and slow otherwise. Salience combines a baseline, decaying endogenous events, and a constant manufactured component:

$$s(t) = \min\left(s_0 + \sum_k m_k 2^{-(t-t_k)/\lambda} \mathbf{1}[t \geq t_k] + c, 1\right) \quad (3)$$

where (t_k, m_k) are event times and magnitudes, λ the half-life in months, and c the manufactured component.

2.4 Trust

The trust threshold is an absolute required excess return, reduced by salience and by discrete validation events:

$$\theta_i(t) = \theta_i (1 - \tau s(t)) \prod_{j:t \geq t_j} (1 - e_j) \quad (4)$$

with τ the salience effect and (t_j, e_j) validation events and their effects. A household adopts only if $g_i - \phi(t)d_i > \theta_i(t)$, where $\phi(t)$ is the prevailing round-trip conversion spread.

2.5 Diffusion

Awareness spreads at different rates within and across the account-holding population. For household i at step t , with $A(t)$ the adopted share of balances and $A_a(t)$ the adopted share among account holders:

$$p_i(t) = \begin{cases} (\alpha + \beta_{\text{in}} A_a(t)) / 12 & i \text{ holds an account} \\ (\frac{1}{2}\alpha + \beta_{\text{ac}} A(t)) / 12 & \text{otherwise} \end{cases} \quad (5)$$

where α is external reach and $\beta_{\text{in}}, \beta_{\text{ac}}$ are the within- and across-population word-of-mouth coefficients. The asymmetry reflects that a listener already holding an account can act immediately, whereas one without must first open an account.

2.6 Allocation

Allocation rises with experience rather than with the size of the benefit. For a household having held for m_i months:

$$w_i(t) = \min\left(w_0 + r \frac{m_i}{12}, \bar{w}, 1 - \min(d_i \kappa / 12, 0.9)\right) \quad (6)$$

with w_0 the initial allocation, r the annual ramp rate, $\bar{w}$ the ceiling and κ the working buffer in months.

This yields a testable implication: mean allocation across adopters remains below the ceiling for as long as adoption is growing, because recent entrants lower the average. In the baseline developed-market run, 48.4 per cent of households have adopted by year ten while mean allocation stands at 39.8 per cent against a ceiling of 60 per cent.

3 Calibration

Diffusion speed is anchored on M-Pesa rather than on cryptocurrency adoption. M-Pesa reached 43 per cent of Kenyan households within seventeen months of its March 2007 launch and approximately 70 per cent within four years. Contemporary accounts attribute this to its agent network, which grew from 4,000 to 19,000 cash-in and cash-out outlets against roughly 850

bank branches nationally, rather than to advertising. The inference drawn is that distribution infrastructure, not awareness, was the binding constraint.

The trust threshold is set low but non-zero. Cohen and Dupas found insecticide-treated bed-net take-up near 75 per cent when free, falling to approximately 20 per cent at a small positive price; a discontinuity at zero rather than an elasticity. A no-premium protective instrument should therefore face a low barrier. It should not face none: a free bed net has no downside, whereas moving savings into a novel instrument carries perceived principal risk.

Salience decay is anchored on the observed pattern in catastrophe insurance, where take-up rises after a nearby event and decays over subsequent years.

4 Data

Quantity	Source
US household savings deposits	Federal Reserve Z.1, B0GZ1FL193030205Q, Q1 2026
UK household deposits	IMF Financial Access Survey, GBRFCLODCHXDC
Japan household cash and deposits	Bank of Japan Flow of Funds, Q1 2026
China household savings	People's Bank of China
India, Turkey, Argentina, Chile	IMF Financial Access Survey
Euro area household deposits	Derived from European Central Bank
Inflation rates	National consumer price indices

Approximately 70 per cent of the savings-pool total is cited. The remainder is derived as nominal GDP multiplied by a household-deposit ratio banded against the two Financial Access Survey ratios retrieved, being India at 39.9 per cent and Turkey at approximately 19 per cent. Brazil, Indonesia and Mexico do not report a household breakdown to the Survey, so complete citation is not attainable from this source.

The denominator is household savings, not M2. M2 includes business deposits, transaction balances and money market fund shares, none of which are addressable; substituting it approximately doubles every figure.

4.1 Account penetration

Published estimates disagree materially, and the access parameter is sensitive to which is adopted. For the United States: Security.org's 2026 consumer report gives 30 per cent of adults from 992 respondents; the National Cryptocurrency Association with Harris Poll gives approximately 25 per cent; Motley Fool Money gives 22 per cent from 2,000 respondents; Pew Research Center gives approximately 20 per cent having ever used, from 8,512 respondents; and the Federal Reserve's Survey of Household Economics and Decisionmaking, from a nationally representative sample of approximately 13,000 adults surveyed in October 2025, gives 10 per cent having used or held within the past year, of which 7 per cent held as an investment.

The Federal Reserve figure is the most methodologically rigorous and is three times lower than the commercial surveys. The discrepancy is partly definitional, since the Survey measures

activity within the past year whereas the commercial surveys measure current ownership, and partly a matter of sampling frames.

What this model requires is neither quantity exactly. Access requires a funded exchange account, which persists after a holder sells, and is therefore closer to having ever owned than to past-year activity. The baseline adopts the upper end on that reasoning; the parameter is swept, and this disagreement is the strongest argument for doing so.

5 Uncertainty

Behavioural parameters have no direct measurement, because no instrument of this kind has been offered to this population. Each is classified as cited, being traceable to a published figure; derived, being computed from cited figures with the calculation stated; anchored, being set against a documented reference class; or unmeasured, having a plausible range only and being swept rather than fixed. No parameter falls outside these categories. The full classification accompanies the code.

5.1 Global sensitivity

Seventeen parameters were sampled simultaneously from continuous ranges across 4,000 draws per market rather than varied individually. One-at-a-time sweeping holds remaining parameters at their defaults and systematically overstates the leading parameter’s influence; an earlier version of this work reported a leverage ratio inflated by that method.

Parameter	Developed	High inflation
Trust threshold	−0.511	−0.132
External reach	+0.287	+0.408
Allocation ceiling	+0.273	+0.379
Account ceiling	+0.255	+0.349
Salience effect on trust	+0.221	+0.140
Account growth	+0.211	+0.320
Allocation ramp rate	+0.197	+0.283
Salience baseline	+0.147	+0.023
Manufactured salience	+0.146	+0.014
Word of mouth, within population	+0.104	+0.143
Round-trip spread	−0.104	−0.041
Churn	−0.097	−0.143
Savings drawdown	−0.078	−0.013
Initial allocation	+0.071	+0.105
Salience half-life	+0.063	+0.056
Adoption hazard	+0.044	+0.060
Word of mouth, across populations	+0.023	+0.028

Table 1: Spearman rank correlation with ten-year share of household savings balances. 4,000 draws per market.

No parameter dominates. The binding constraint differs by market: trust in developed economies, at approximately twice the influence of any other parameter, and awareness where inflation is high. Manufactured salience registers in developed markets and is negligible in high-inflation

ones, consistent with a campaign being unable to add vividness to a threat already experienced directly.

5.2 Outcome distribution

	Developed	High inflation
Minimum	0.04%	0.99%
5th percentile	1.12%	3.87%
25th percentile	6.13%	8.09%
Median	13.21%	12.54%
75th percentile	21.63%	18.31%
95th percentile	37.82%	30.63%
Maximum	73.87%	58.04%
Samples with adoption	99.92%	100.00%
Samples self-sustaining at year ten	92.0%	98.8%

Table 2: Distribution of ten-year outcomes as a share of household savings balances, across 4,000 draws per market.

Across 8,000 samples spanning seventeen simultaneously varied unmeasured parameters, adoption occurred in all but three.

6 Reproducibility

A run is fully determined by its configuration file and seed. No global random number generator, no wall-clock time and no environment variable enters the model. Each run writes a manifest recording the configuration hash, seed, commit hash, library versions and a checksum of every output. Reference fixtures are committed, and the test suite fails where a differing environment produces differing figures. Code, configurations and the parameter register are published so results may be reproduced, modified or contested.

7 Limitations

Categorical access barriers are treated as scenario boundaries. Markets whose capital controls prohibit the instrument are excluded from the base case and reported separately, in the manner of an epidemiological study excluding a population to which an intervention cannot lawfully be offered. Jurisdiction-specific compliance questions are out of scope as implementation-dependent.

China is reported separately and is never included in an aggregate total. At approximately \$22.5 trillion it is the largest single pool and would dominate any aggregate containing it.

Early-year figures carry the widest uncertainty. The initial segment of a diffusion curve is governed almost entirely by unmeasured parameters; small changes move the first year by multiples while barely affecting the tenth.

Terminal figures depend heavily on two unmeasured ceilings, being allocation and account penetration, each swept but neither measured. Account penetration estimates vary threefold

across published sources; a reader preferring the Federal Reserve figure should read the lower tail of the distribution rather than the median.

No competitive or regulatory response is modelled.

8 Corrections to earlier versions

Four findings reported during development were subsequently identified as artefacts rather than results.

A reported source-pool asymmetry of between 4:1 and 47:1 between local-currency holders and stablecoin holders arose from an error in the friction calculation, which annualised conversion cost by transaction count rather than by the portion of the balance withdrawn, rendering the stablecoin cohort's net advantage negative. Corrected, both cohorts adopt.

A reported allocation asymmetry of 3.75 was the ratio of two configuration values set by hand, being an input restated rather than a model output. Across a range moving that ratio fivefold, the ten-year result varied by 17 per cent.

A reported awareness threshold, below which adoption fails to become self-sustaining, is a property of the diffusion form rather than a feature of the instrument. Its location moves with the word-of-mouth coefficient and disappears at high values.

A reported parameter leverage ratio of 13.7 was produced by one-at-a-time sweeping. Under simultaneous sampling no parameter dominates.

9 Measurements that would improve this work

A landing-page conversion test in two target markets with real product terms, which would measure the developed-market binding constraint directly. A general-population survey screened for savings balance, being the only route to demand data uncontaminated by cryptocurrency-user sampling. Household penetration data in place of transaction-volume growth for diffusion calibration. Market-maker spread indications, once a venue exists.

Nothing in this document constitutes an offer to sell or a solicitation of an offer to buy any security or financial instrument, nor investment, financial, legal or tax advice. Outputs are conditional projections derived from stated assumptions, many of them unmeasured. They are not forecasts.